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1.0 Specific Information:

No.	Unit No.	Serial Number	TC	Test Requested
1.	MAIN TRAFO UNIT 1	1006901	4	OLTT+ and Corrosive Sulfur
2.	TRAFO 1 ESP UNIT 1	10937	5	OLTT+ and Corrosive Sulfur
3.	TRAFO 2 ESP UNIT 1	10035	6	OLTT+ and Corrosive Sulfur
4.	TRAFO 3 ESP UNIT 1	10031	7	OLTT+ and Corrosive Sulfur
5.	TRAFO 4 ESP UNIT 1	10034	8	OLTT+ and Corrosive Sulfur
6.	MAIN TRAFO UNIT 3	1122170	9	OLTT+ and Corrosive Sulfur
7.	UAT 3	1035082	10	OLTT+ and Corrosive Sulfur
8.	TRAFO 1 ESP UNIT 3	007308428	11	OLTT+ and Corrosive Sulfur
9.	TRAFO 2 ESP UNIT 3	0070084208	12	OLTT+ and Corrosive Sulfur
10.	TRAFO 3 ESP UNIT 3	0072084208	13	OLTT+ and Corrosive Sulfur
11.	TRAFO 4 UNIT 3	0071084208	14	OLTT+ and Corrosive Sulfur
12.	ASH HANDLING	10820108	15	OLTT+ and Corrosive Sulfur
13.	ASH HANDLING	10820107	16	OLTT+ and Corrosive Sulfur

Note:

* OLTT+ consists of Liquid Screen Tests, Dissolved Gas Analysis (DGA), Water Content, Furan Analysis, Metals-in-Oil Analysis, Liquid Power Factors, and Inhibitor Content (DBPC).

** Liquid Screen Test consists of Acidity, Interfacial Tension (IFT), Dielectric, Colour, Specific Gravity, Visualization, and Sediment.

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2.0 Our Comments and Recommendations of The Test Results Analysis:

We had reviewed each laboratory test result against respective benchmark parameter and following are our comments and recommendations.

No.	Unit No.	Serial Number	TC
1.	MAIN TRAFO UNIT 1	1006901	4

A) Analysis

- **Acidity** of 0.017mgKOH/g and **IFT** of 41.5mN/m are **ACCEPTABLE**.
- **Dielectric** of 58kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **DGA** shows significant amount of Hydrogen coupled with Carbon Monoxide and Carbon Dioxide. High Hydrogen and Carbon Oxides in a transformer's oil is an indication that the transformer is afflicted with a partial discharge condition. It is advisable to consult transformer manufacturer to ascertain the cause of fault gases buildup in the transformer.
- **Water Content** in oil of 10ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.82% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.24% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

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B) Recommendation

- Retest for **DGA** in 6 months to diagnose any possible fault and establish trend of the gases.
- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
2.	TRAFO 1 ESP UNIT 1	10937	5

A) Analysis

- **Acidity** of 0.056mgKOH/g is **QUESTIONABLE** whereas **IFT** of 25.3mN/m is **UNACCEPTABLE**. This indicates the oxidation process has begun in the unit. Oxidation process produces acid and sludge, which contaminate the oil and the core, thus causes degradation of the paper insulation, as well as reduces the efficiency of heat dissipation.
- **Dielectric** of 50kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 18ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 2.34% M/dw is graded as **D**. For this transformer, it is advisable to maintain the water content level below 30ppm.
- **Furanic compound** 2FOL of 95ppb was detected which indicates the solid insulation is wet. It is important to maintain the oil in a good condition so as not to further deteriorate the paper insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur test** on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Most of the parameters show that the liquid insulation is in bad condition. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
3.	TRAFO 2 ESP UNIT 1	10035	6

A) Analysis

- **Acidity** of 0.042mgKOH/g is **ACCEPTABLE** while **IFT** of 26.6mN/m is **UNACCEPTABLE**. Although the Acidity level is acceptable, it is at the borderline of acceptable and questionable range. Together with IFT, these two parameters indicate the presence of oxidation byproducts in the oil. Oil oxidation byproducts promotes more acids to be produced by the present acids. Therefore, removal of these contaminations is highly recommended.
- **Dielectric** of 41kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 21ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 2.48% M/dw is graded as **D**. For this transformer, it is advisable to maintain the water content level below 30ppm.
- **Furanic compound** 2FOL of 62ppb was detected which indicates the solid insulation is wet. It is important to maintain the oil in a good condition so as not to further deteriorate the paper insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.02% by weight is detected, thus **UNACCEPTABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in light orange, thus **Non-corrosive of 1A**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Most of the parameters show that the liquid insulation is in bad condition. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
4.	TRAFO 3 ESP UNIT 1	10031	7

A) Analysis

- **Acidity** of 0.048mgKOH/gm is **ACCEPTABLE**, while **IFT** of 28.4mN/m is **QUESTIONABLE**. These two parameters indicates the starting formation of oxidation by products in the oil. Close monitoring is advised.
- **Dielectric** of 44kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA:** All of the **gases** are assessed as acceptable and indicates no problems with fault condition for this transformer.
- **Water content** in oil of 32ppm is **QUESTIONABLE**. Meanwhile Water Content in solid insulation of 4.16% M/dw is graded as **E**. For this transformer, it is advisable to maintain the water content level below 30ppm. This indicates the presence of high water content in the oil.
- **Furanic compound** 2FOL of 32ppb was detected which indicates the solid insulation is wet. It is important to maintain the oil in a good condition so as not to further deteriorate the paper insulation.
- **Metals-In-Oil** analysis shows the amount of dissolved metals detected is below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.01% by weight is detected, thus **UNACCEPTABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in light orange, thus **Non-corrosive of 1A**.

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B) Recommendation

- Retest **Acidity, Interfacial Tension, and Liquid Power Factor** in 6 months to monitor their levels.
- Retest **Dielectric** and **Water Content** in 6 months to assess if the insulating oil deteriorate to a level that require corrective action to improve the oil quality.
- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
5.	TRAFO 4 ESP UNIT 1	10034	8

A) Analysis

- **Acidity** of 0.028mgKOH/g is **ACCEPTABLE** while **IFT** of 26.3mN/m is **UNACCEPTABLE**. The low IFT level indicates the oil is containing contaminants that had weakened the Interfacial tension of the oil. Removal of these contaminations is highly recommended.
- **Dielectric** of 24kV is **UNACCEPTABLE**. This is reflected from the high moisture content detected in the transformer oil.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 29ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 3.77% M/dw is graded as **E**. The value is approaching the borderline of questionable level. For this transformer, it is advisable to maintain the water content level below 30ppm.
- **Furanic compound** 2FOL of 18ppb was detected which indicates the solid insulation is wet. It is important to maintain the oil in a good condition so as not to further deteriorate the paper insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factor** for 25°C is **QUESTIONABLE**, while at 100°C is **UNACCEPTABLE**. This indicates the presence of contaminant particles in the insulating oil. Aging and oxidation of the oil may have contributed to the high liquid power factor.
- **Oil Inhibitor** of 0.08% by weight is detected, thus **UNACCEPTABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation.
- Tested on **copper** strip resultant in multicolored with lavender blue and/or silver overlaid on claret red, thus **Non-corrosive of 2C**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit.
Most of the parameters show that the liquid insulation is in bad condition. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.
- Kindly perform checks on the Silica Gel. It is found to be in bad condition during sampling by our technician. Replace silica gel if it is confirmed to be in bad condition to prevent moisture from entering into the transformer.

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No.	Unit No.	Serial Number	TC
6.	MAIN TRAFO UNIT 3	1122170	9

A) Analysis

- **Acidity** of 0.017mgKOH/g and **IFT** of 42.3mN/m are **ACCEPTABLE**.
- **Dielectric** of 78kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- The **DGA** result shows that Carbon Monoxide level is high. Other gases remain within normal limits and does not indicate any fault afflicting the transformer unit. Proceed with annual monitoring.
- **Water Content** in oil of 15ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.23% M/dw is graded as **B**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.22% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
7.	UAT 3	1035082	10

A) Analysis

- **Acidity** of 0.022mgKOH/g and **IFT** of 38.1mN/m are **ACCEPTABLE**.
- **Dielectric** of 60kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- The **DGA** result shows that Carbon Monoxide and Carbon Dioxide levels are high. Other gases remain within normal limits and does not indicate any fault afflicting the transformer unit. Proceed with annual monitoring.
- **Water Content** in oil of 11ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.90% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.23% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Kindly perform checks on the Silica Gel. It is found to be in bad condition during sampling by our technician. Replace silica gel if it is confirmed to be in bad condition to prevent moisture from entering into the transformer.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
8.	TRAFO 1 ESP UNIT 3	007308428	11

A) Analysis

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE** while **IFT** of 23.2mN/m is **UNACCEPTABLE**. The low IFT level indicates the oil is containing contaminants that had weakened the Interfacial tension of the oil. Removal of these contaminations is highly recommended.
- **Dielectric** of 32kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA:** All of the **gases** are assessed as acceptable and indicates no problems with fault condition for this transformer.
- **Water content** in oil of 26ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.85% M/dw is graded as **D**. For this transformer, it is advisable to maintain the water content level below 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation.
- **Corrosive sulfur test** on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
9.	TRAFO 2 ESP UNIT 3	0070084208	12

A) Analysis

- **Acidity** of 0.118mgKOH/g and **IFT** of 16.2mN/m are **UNACCEPTABLE**. At this level of Acidity and IFT, oxidation process are occurring at an increasing rate. Visible sludge is already formed in and around the transformer core. These oxidation products are in fact attacking the paper insulation and tearing them apart. Loss of the paper's mechanical and dielectric strength signals the end of the useful life of the transformer.
- **Dielectric** of 24kV is **UNACCEPTABLE**. This is reflected from the high moisture content detected in the transformer oil.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water Content** in oil of 39ppm is **UNACCEPTABLE**. Meanwhile Water Content in solid insulation of 2.54% M/dw is graded as **D**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm. This indicates the presence of high water content in the oil. The presence of high water content could cause dielectric failure.
- **Furanic compound** 2FAL of 169 ppb was detected which indicates normal aging of the solid insulation.
- **Metals-In-Oil** analysis shows the amount of dissolved metals detected is below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- Tested on **copper** strip resultant in multicolored with lavender blue and/or silver overlaid on claret red, thus **Non-corrosive of 2C**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit to improve Acidity, IFT, Dielectric, and Water Content. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as excess moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
10.	TRAFO 3 ESP UNIT 3	0072084208	13

A) Analysis

- **Acidity** of 0.025mgKOH/g is **ACCEPTABLE** while **IFT** of 27.8mN/m is **UNACCEPTABLE**. The low IFT level indicates the oil is containing contaminants that had weakened the Interfacial tension of the oil. Removal of these contaminations is highly recommended.
- **Dielectric** of 40kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA:** All of the **gases** are assessed as acceptable and indicates no problems with fault condition for this transformer.
- **Water content** in oil of 27ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.75% M/dw is graded as **D**. For this transformer, it is advisable to maintain the water content level below 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.01% by weight is detected, thus **UNACCEPTABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation.
- Tested on **copper** strip resultant in multicolored with lavender blue and/or silver overlaid on claret red, thus **Non-corrosive of 2C**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
11.	TRAFO 4 UNIT 3	0071084208	14

A) Analysis

- **Acidity** of 0.020mgKOH/gm is **ACCEPTABLE**, while **IFT** of 31.2mN/m is **QUESTIONABLE**. This indicates the starting formation of oxidation by products in the oil.
- **Dielectric** of 62kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA:** All of the **gases** are assessed as acceptable and indicates no problems with fault condition for this transformer.
- **Water content** in oil of 19ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.80% M/dw is graded as **D**. For this transformer, it is advisable to maintain the water content level below 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** were not detected, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.01% by weight is detected, thus **UNACCEPTABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

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B) Recommendation

- Retest **Acidity, Interfacial Tension, and Liquid Power Factor** in 6 months to assess if the insulating oil deteriorate to a level that require corrective action to improve the oil quality
- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
12.	ASH HANDLING	10820108	15

A) Analysis

- **Acidity** of 0.022mgKOH/g and **IFT** of 38.1mN/m are **ACCEPTABLE**.
- **Dielectric** of 38kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- The **DGA** result shows that Carbon Monoxide and Carbon Dioxide level are high. Other gases remain within normal limits and does not indicate any fault afflicting the transformer unit. Proceed with annual monitoring.
- **Water Content** in oil of 28ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 2.30% M/dw is graded as **C**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.23% by weight is detected, thus **ACCEPTABLE**.
- **Corrosive sulfur test** on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
13.	ASH HANDLING	10820107	16

A) Analysis

- **Acidity** of 0.022mgKOH/g and **IFT** of 37.3mN/m are **ACCEPTABLE**.
- **Dielectric** of 36kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water Content** in oil of 21ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 2.20% M/dw is graded as **C**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.22% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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3.0 Summary of Our Comments & Recommendations:

No.	Sub Name	Unit No.	TC	Recommendation
1.	MAIN TRAFO UNIT 1	1006901	4	• Retest 6 months: DGA • Add inhibitor 2,6-DBPC and passivator • Other parameters: Periodic condition monitoring
2.	TRAFO 1 ESP UNIT 1	10937	5	• Perform OIL REGENERATION
3.	TRAFO 2 ESP UNIT 1	10035	6	• Perform OIL REGENERATION
4.	TRAFO 3 ESP UNIT 1	10031	7	• Retest 6 months: Acidity, IFT, Liquid Power Factor, Dielectric, and Water Content • Add inhibitor 2,6-DBPC and passivator • Other parameters: Periodic condition monitoring
5.	TRAFO 4 ESP UNIT 1	10034	8	• Perform OIL REGENERATION • Inspect silica gel
6.	MAIN TRAFO UNIT 3	1122170	9	• Add passivator • Periodic condition monitoring
7.	UAT 3	1035082	10	• Add passivator • Periodic condition monitoring • Inspect silica gel
8.	TRAFO 1 ESP UNIT 3	007308428	11	• Perform OIL REGENERATION
9.	TRAFO 2 ESP UNIT 3	0070084208	12	• Perform OIL REGENERATION
10.	TRAFO 3 ESP UNIT 3	0072084208	13	• Perform OIL REGENERATION

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No.	Sub Name	Unit No.	TC	Recommendation
11.	TRAFO 4 UNIT 3	0071084208	14	• Retest 6 months: Acidity, IFT, and Liquid Power Factor • Add inhibitor 2,6-DBPC and passivator • Other parameters: Periodic condition monitoring
12.	ASH HANDLING	10820108	15	• Add passivator • Periodic condition monitoring
13.	ASH HANDLING	10820107	16	• Add passivator • Periodic condition monitoring

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